

Kamyaab — Backend Development Roadmap (Updated)

7-week plan — updated to align with the confirmed frontend roadmap and the official pitch presentation. Team: Momna, Abdullah, Wajid — shared backend ownership.

Changelog — What Changed From the Original Roadmap

1. Voice AI classification REMOVED from Week 6 — confirmed Phase 2 per pitch deck, not MVP. 2. Two-way ratings ADDED to Week 6 — worker rates client too, not just client rates worker. 3. Reference check ADDED to Week 3 worker verification, alongside CNIC. 4. Full worker-category seed list ADDED to Week 1 (was sample-only). 5. Priority listing (paid visibility) ADDED to Week 5 matching/ranking. 6. My Jobs history/list endpoint ADDED to Week 4. 7. Auth confirmed as Supabase Auth (no change — this was already correct). 8. Request→accept→then-contact flow CONFIRMED correct (matches pitch deck's core trust model) — no change needed to Weeks 4–5 booking logic.

Tech Stack

Layer	Choice	Notes
Backend runtime	Node.js (v20+) + TypeScript	Type safety across the codebase
Web framework	Express	Minimal, transparent, easy to debug
Database	PostgreSQL via Supabase	Hosted DB, no server management
ORM	Prisma	Type-safe queries, migrations
Authentication	Supabase Auth	Confirmed — phone OTP (clients), email/password (agents/admin)
File storage	Supabase Storage	Private buckets, signed URLs for CNIC etc.
Background jobs	DB-backed queue	Polling — no Redis needed yet
SMS	Mock (Weeks 1–7)	Logs actions instead of a real provider — no change
Voice AI	Phase 2 — removed from MVP	Confirmed via pitch deck: voice + AI matching is Phase 2, not MVP

Open item (unchanged): CNIC number field encryption approach still awaiting a team decision. Treat as sensitive until resolved.

Week 1 — Project Skeleton & Database Schema

Goal: A running Express server connected to the database, with the first migration and full seed data in place.

- Set up the project: Express, TypeScript, Prisma, tsconfig.

- Minimal Express server with health-check route (GET /ping).
- npx prisma init, connect to Supabase DB via .env.
- Define core schema: User, ClientProfile, WorkerProfile, Category, Area (Karachi zones, parent/child).
- Run first migration: npx prisma migrate dev --name init.
- **UPDATED: seed the FULL category list — electrician, plumber, carpenter, gardener, painter, driver, tailor, cook (was sample-only before).**
- Draft API contract stubs for POST /auth/otp/request and POST /auth/otp/verify.

Week 2 — Authentication & Worker Onboarding

Goal: Clients log in via phone OTP; agents create worker profiles pending approval. (No change from original — already correct.)

- Phone OTP flow via Supabase Auth (signInWithOtp / verifyOtp) for clients.
- authenticate middleware verifies JWT, attaches user ID + role.
- Auto-create ClientProfile on first login.
- Agent/admin login via email + password (signInWithPassword); seed fixed accounts.
- POST /api/v1/workers (agent-only): creates WorkerProfile, status = PENDING_APPROVAL.
- PATCH /api/v1/workers/:id/verify (admin-only): approve or suspend.

Week 3 — Worker Profiles, Verification & Search

Goal: Clients browse approved workers filtered by category and area.

- Expand WorkerProfile: cnicNumber (sensitive), cnicFrontPath, cnicBackPath, rating, completedJobsCount.
- **UPDATED: add referenceCheck fields (referenceName, referencePhone, referenceVerifiedBy, referenceVerifiedAt) — pitch deck defines "verified" as CNIC + reference check, not CNIC alone.**
- Add verifiedBadges JSON field (identity checked, phone confirmed, reference checked).
- CNIC document upload: POST /api/v1/workers/:id/documents — private Supabase Storage bucket.
- Public search endpoint: GET /api/v1/workers — filtered by category/area, APPROVED only, excludes phone/address/CNIC.
- Basic ranking: order by rating, then completed jobs count.

Week 4 — Job Requests & Booking Flow (Specific Worker)

Goal: A client requests a specific worker; on acceptance, a booking is created and contact is released. Flow confirmed correct against the pitch deck's trust model — no change to the core logic.

- JobRequest model: client, category, area, description, urgency, budget, type (SPECIFIC_WORKER), target worker, status.
- POST /api/v1/job-requests (draft) and POST /api/v1/job-requests/:id/submit (validates, moves to WORKER_CONTACTED).
- Invitation model + generate one invitation per submitted specific-worker job.
- Mock SMS on invitation sent.
- POST /api/v1/invitations/:id/accept: creates Booking, releases contact info to both parties, mock SMS confirmation.
- Expiry logic: background job expiring invitations (and parent job) after 24 hours.
- **UPDATED: add GET /api/v1/job-requests/mine — lists the logged-in client's own job requests/bookings with status, for the frontend's My Jobs screen (was missing entirely before).**

Week 5 — Open Jobs, Availability, Priority Listing & Admin Override

Goal: Clients post open jobs matched to multiple workers; workers set availability; paid priority listing affects ranking; admins override manually.

- Extend JobRequest: type = OPEN, moves to MATCHING on submission.
- Deterministic matching function: filter by category/area/availability, rank by availability, rating, completed jobs — NOT AI-based (pitch deck confirms AI matching is Phase 2).
- **UPDATED: add isPriorityListed boolean (or similar) to WorkerProfile — priority-listed workers rank above equivalent non-priority workers in matching and search results. This is the paid "Priority listing" revenue stream from the pitch deck — no payment processing needed yet, just the boolean flag and ranking boost.**
- Batch-create invitations for matched workers, trigger mock SMS for each.
- Simple worker-facing web page: OTP login, list of pending invitations, accept/reject.
- WorkerAvailability model (available/busy/unavailable/unknown), update endpoint, auto-expiry after 24 hours.
- POST /api/v1/admin/assign-worker for manual admin assignment.

Week 6 — Reviews, Complaints & Audit Trail

Goal: Reviews, complaints, and audit logging are added. Voice AI is NOT part of this week — moved to Phase 2 per pitch deck.

- **REMOVED: real speech-to-text/LLM voice classification integration — this is now Phase 2, not MVP. Do not build POST /api/v1/voice-jobs/classify this cycle.**
- Build reviews: POST /api/v1/bookings/:id/reviews, only for completed bookings; recalculates worker's average rating.
- **UPDATED: two-way reviews — add POST /api/v1/bookings/:id/reviews/worker-to-client (or a role field on the same endpoint) so workers can rate clients too, per pitch deck's "accountability on both sides."**
- Build complaints: POST /api/v1/bookings/:id/complaints with description + evidence files; admin can view/resolve.
- AuditLog table + middleware recording every status-changing action (actor, action, before/after).

Week 7 — Integration Testing, Security & Staging Deploy

Goal: Confirm everything works together and ship a working staging environment. (No change from original.)

- Integration tests covering core journeys (agent creates worker → admin approves → client searches → books → completes → reviews both ways).
- Security pass: CNIC images only via signed URLs, client addresses never in public search, OTP endpoints rate-limited.
- Deploy to staging (Render/Railway), connected to a separate/test Supabase project.
- Manually run full acceptance journeys on staging.
- Write README.md: local setup, env variables, seeding instructions.

Build Order — Summary

#	What
1	Express server + Prisma connection
2	Database tables + FULL category/area seed data

3	Phone OTP login (clients) via Supabase Auth
4	Agent + admin login (email/password)
5	Worker CRUD incl. CNIC + reference check (agent creates, admin approves)
6	Worker search (public, filtered by category/area)
7	Specific-worker request → invitation → accept → booking
8	My Jobs / job-history endpoint
9	Open-job request → deterministic matching → batch invitations → accept → booking
10	Worker availability + priority listing (paid ranking boost)
11	Admin manual assignment
12	Two-way reviews + complaints
13	Audit logging
14	Testing + staging deployment

Explicitly Out of Scope for This Cycle (Phase 2+)

- Voice-based job posting & AI-assisted matching — Phase 2 per pitch deck.
- Material/supplier bundling — Phase 2 per pitch deck.
- B2B housing-society partnerships — Phase 2 per pitch deck.
- Multi-city expansion — Phase 3 per pitch deck.
- Real payments and real SMS — mocked for MVP, unchanged.

Still Open

- CNIC encryption approach — awaiting team decision, unchanged from original roadmap.